

Keeping up with the digital natives: Using social media in an all-girls science classroom

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Abstract

This qualitative, ethnographic study focuses on how combining aspects of visual learning with a social media application (app)—Instagram—can be a useful learning tool for digital natives (DNs). Current students are classified as digital natives because they are inundated with technology, which has become an essential tool in their everyday lives. By combining learning with a social media app that relies heavily on virtual communication, a new educational tool can emerge that is both engaging and relevant. Using its Live feature, students incorporate an app they use regularly in their nonacademic lives and bring it into the educational setting of the science classroom. Thirty-two female, high school biology students received permission to respond to a questionnaire and participate in a focus group or individual interview to further discuss their reactions to use Instagram in the biology classroom. Four students were asked additional questions regarding their reactions and understandings of how Instagram Live was used and how it enhanced their biology content understanding. The themes that emerged considered largely how Instagram as an educational tool brings science into the daily lives of digital natives. Implications from the study show how the use of Instagram specifically supported young women in science.

KEYWORDS

equity, science/science education, social media, teachers and teaching

1 | INTRODUCTION

Digital natives (DNs) are individuals born and raised in a world in which technology is widely used and heavily relied upon (Palfrey & Gasser, 2008). This current generation of high school students is classified as DN because they live in a digital world and are inundated with technology as an essential tool in their everyday lives. DNs rely on constant access to digital technologies such as tablets, smartphones, smartwatches, and laptops that provide them with instant access to unlimited information and communication capabilities. “Major aspects of their lives—social interactions, friendships, civic activities—are mediated by digital technologies. And they’ve never known any other way of life” (p.

iv). As we consider DNs as a unique generation that heavily relies on digital technologies, it is important to consider the learning styles employed by this generation and how they may differ from those of the past. Palfrey and Gasser go on to describe that DNs are creative, quick to adapt and learn how to use new software, and “rely on this connected space for virtually all of the information they need to live their lives” (p. x). DNs are transforming how information is perceived, transmitted, and shared (Adobe Communications Team, 2018; Roe, 2019).

Digital information sharing among the DNs has an impact on education. “Because of their interest in working in groups and their low tolerance for boredom, the traditional lecture may not be as effective ... as it was with previous

generations” (Roehling et al., 2011, p. 1). DNs require a classroom experience that engages them, not one that talks at them. With access to multiple forms of technology, with social media being a common one, DNs and their use of social networking technology tools have been embraced and used variably. Therefore, the impact of social media will continue to advance and become an even greater part of the lives of these DNs, especially as they move into the workforce in greater numbers, achieve leadership and management positions, and raise families. In all these areas, the use of social media and technology tools will undoubtedly continue to be part of their daily life (Stephens & Gunther, 2016). With such an enormous impact and exposure, using social media tools in the academic lives of students seem obvious for teachers to engage DNs and to use these tools in the classroom.

Social media encompasses various technological tools that allow for networking, content-sharing, and interaction using digital tools. Social media has become an integral part of the daily life of many individuals in the United States (Burke & Foulger, 2014). It allows individuals to communicate with each other regardless of location; all that is required is a form of technology, such as a smartphone or a computer, and an internet connection. Moreover, social media is comprised of many apps that facilitate communication in various formats. This research centers around one particular social media platform, Instagram, and its feature Instagram Live, as a social media learning tool in a secondary biology classroom.

2 | PURPOSE AND RESEARCH QUESTION

Social media is continually evolving, and with that, so is the digital literacy capability of our students. Communication, privacy, and the need to share the many aspects of daily life have become more common in the lives of youth than in the past decade. Knowing how to harness these issues and use them to address digital literacy in the classroom is key. Vasudevan (2010) stated, “By paying attention to digital geographies, particularly the navigation across digital spaces and orchestration of multiple modalities, educators can cultivate youths’ literacies while at the same time inspire new sites of education” (p. 79). Educators must find ways to implement student social media use appropriately and effectively in and out of school because our DNs both “want—and need—new structures for learning, and these pervasive demands are changing the game” (Adobe Communications Team, 2018, para 10). This is particularly important because we want to improve and have better communication methods, but with so many choices, new tools, and strategies, it may be more challenging to harness the good of these digital tools for educational use in classrooms (Jones-Kavalier & Flannigan, 2006). Using social media and conducting

research on using social media is a relatively new field of study. Thus, the approach we took highlights the potential of social media as a learning tool in K–12 science classrooms.

This study explored the student experience by using Instagram and its features as a social media learning tool in the science classroom. By hosting an Instagram Live science review session, students were able to access their classroom teacher and their peers from any location with cell phone service or Wi-Fi initiating an investigation into the potential that social media can have in the science classroom and how it can increase the reach of academic communication outside the brick and mortar school structure. The following research question guiding this study was: How does the implementation of Instagram as a social media tool engage student learning in an all-female, secondary biology classroom?

3 | LITERATURE REVIEW

3.1 | Social media and youth culture

According to information gathered from Pew Studies on how teens engage with social media, Auxier (2020a) found that 72% of teens use Instagram and it “ranks as the second most popular platform among Americans in this age group” (para 6). Social media should be leveraged as a pedagogical tool to be a classroom connector between course content and the visual and digital literacy of the DN student (Auxier, 2020b; Carpenter et al., 2020; Stephens & Gunther, 2016). Today’s teachers must learn to communicate in the language and style of their students. This does not mean changing the meaning of what is important, or not encouraging critical thinking skills. But it does mean adjusting the pace and presenting information in ways that are more in line with the interests and values of DNs. The integration of social media in the classroom provides a supplementary benefit to make learning culturally responsive (Gay, 2018) by integrating aspects of current youth culture. By recognizing that social media is very much an essential part of the lives and social culture of our students, we can use these digital tools in our classroom to incorporate youth culture.

Senge et al. (2012) discuss learning communities as those heavily influenced by the social constructs that create them. They discuss the idea that social media has enabled a new age of culture and learning to co-exist by developing new relationships with their peers, teachers, and content in and out of the classroom. “Attention to digital geographies, and attendant communicative practices, can yield important insights about education beyond the school walls, which can inform how we construct spaces for education and literacy practices within schools” (Vasudevan, 2010, p. 62). The authors looked to recognize the possible ways in which Instagram provided an educational opportunity in the learning of science while

still using authentic, scientific language, diagrams, and models to convey biological concepts. To find ways to integrate Instagram into the science classroom, one must understand its features, functions, and how it is used by our students in their social cultures. Technology and social media enable collaboration, as well as conversation and learning (Blair & Serafini, 2014) which can be achieved by posting images, text, and video on Instagram.

Because learning happens contextually, educators need to find ways youth can learn in a familiar and accessible way (Jordan, 2010). Instagram is very much a cultural experience for young people today. Social media has become a part of the daily culture as a main channel for communication and collaboration among individuals. “By linking students’ learning experiences outside of educational institutions to those within the institutions’ contexts, students’ learning could be enhanced, and institutions should be supportive of this” (Ng, 2012, p. 6). Because learning is also very much a part of the culture of students, implementing social media, which many students check habitually throughout the day, facilitates learning as a part of their daily routine via digital tools. “Pedagogically speaking, mobile technologies have the potential to support a number of learning theories, including behaviorist (e.g., student response system), constructivist (e.g., gaming), situated (e.g., authentic learning from museums, etc.), and collaborative learning (e.g., communication tools)” (Thomas & O’Bannon, 2013, pp. 12–13).

4 | CONCEPTUAL FRAMEWORK

To guide this study, a conceptual framework for digital learning was used. Hadjerrouit (2010) developed a conceptual framework for using and evaluating web-based learning resources (WBLR) in schools. The framework for education proposes that WBLR should have technical usability, pedagogical usability, and appropriate content. In the way that Instagram was used in this study, it met each of these criteria. Instagram is a technical platform that serves as an interactive, goal-oriented, autonomous learning environment. It meets the needs of the students pedagogically while conveying relevant content to our digitally literate youth culture.

4.1 | Digital literacy and learning

Recognizing that digital communication holds power, particularly for a generation that has a strong relationship with digital devices and social media platforms, educators need to expand their methods of teaching in and out of the classroom to address the use of digital technologies and the development of digital literacy. “Experts in the field suggest that the current generation of teenagers - sometimes referred to as the

E-Generation - possesses digital competencies to effectively navigate the multidimensional and fast-paced digital environment” (Jones-Kavalier & Flannigan, 2006, p. 8). These “digital competencies” include, but are not limited to email, social media, and digital notetaking. Additionally, students are currently receiving much of their news, interactions, and information in a digital format. We define the possession of these digital competencies as being digitally literate, but is this fluency in digital media being integrated into the classroom to its full potential? By dismissing digital literacy as being separate from the classroom and students’ lives, we ignore its usefulness to promote and combine critical thinking, visual literacy, and technological skills; we neglect to engage youth culture in the classroom. Digital literacy must be integrated with the content and curricula that permeates our educational system. “These days, new media literacy technical skills catapult traditional learning methods into orbit” and the “traditional chalkboards and overheads with pens do not occupy the same realm as current capabilities” (Jones-Kavalier & Flannigan, 2006, p. 9).

5 | METHODOLOGY

5.1 | Field setting and participants

The study was conducted at the high school at which the first author is currently employed. It is a small, metropolitan, Catholic college preparatory school in New York City. Students range in age from 13 to 18 years old and all had access to technology, at school and at home. With many students coming from socioeconomic backgrounds, all receive a partial scholarship to attend the school, in addition to some students receiving either full or partial merit scholarships. The school, although private, offers a competitively priced tuition, significantly lower than many others in the surrounding area. Since the entire population of 250 students is female, the study focuses on female participation and their use of Instagram in learning science. The first author acted as both the educator and the principal researcher of this study.

The sampling method was out of convenience (Merriam & Tisdell, 2016), as students were current or previous students in the biology classroom and had used Instagram as a learning tool with first author. The students were all aware of Instagram as it was a popularly used app among the students at this school; therefore, this group of students was already familiar with and practicing its uses and features in their social lives. Students were instructed that refusal to participate or removal from the study would not impact their course grades or admission to future science classes. Students interested in participating in the study were given both Assent and Consent Forms before the study began. Because the students were of high school age, they needed parental permission in addition

to completing the assent form. Students were provided with an identification code upon return of their consent and assent forms. To maintain confidentiality, all students were given a pseudonym as well. In total, 32 students (ages 13 to 17) opted to participate in the data collection process; 15 students self-identified as White, 4 as Asian American, 7 as Hispanic, 2 as African American, and 4 identified as multiple categories; and 10 students were in Advanced Placement (AP) Biology and 22 were in 9th grade Honors Biology.

5.2 | Procedures

5.2.1 | Setting up Instagram

A single Instagram profile was created for communication purposes between the first author and her Biology and Advanced Placement Biology courses. She had used Instagram in the biology classroom for the past three years. The motivation behind using Instagram with her students began when the app was first “trending” or becoming popular as a social media platform and widely used by her students. After some requests from students to use Instagram in the class, she decided to try integrating it into her course as an extracurricular resource for students. Initially, the Instagram account was used to post homework reminders, lab pictures, biology memes that use humor to relate to the course material, current events, and discoveries. These varied usages allowed the students to be exposed to science in a different or “non-traditional” way. The girls in both the AP and the Honors classes were encouraged to follow Instagram at the start of the school year because a stream of images and information related to current class topics would be posted throughout the year. Students who were not allowed to have an account or did not have a cell phone were able to access the account by using the free Instagram website. For that reason, the Instagram account was public, so it could easily be accessible at any time from the app or the website by the students. The school had available desktop computers throughout the building which students could use to access Instagram also.

5.2.2 | Instagram live review session

One Instagram Live event was held mid-year (January) of the study. Because the Live session took place at 7 p.m., everyone accessed the stream from locations outside the school building. The session was recorded with a computer camera so students could access the conversation later as a saved video on Google Classroom, another social and educational learning platform. Instagram has since changed the capability to save Live sessions so that sessions can be saved directly through the Instagram app. The Live stream session

was held to help students review for their biology midterm, which was given the following day in class. Students had requested holding the Instagram Live as a session for review. Students without an Instagram account were not able to access the Live conversation, but they could view the saved recording that was created and posted immediately following the session. Some girls shared their “Finsta” accounts (a fake or additional Instagram account) with other students to increase access and participation in the Live review session.

Students joining the Live review session were able to type questions that could be read by all participants from the screen of their computer, cell phone, or other devices. As the questions appeared, the teacher was able to address each one and answer them using verbal responses as well as gestures that students could see. Students asked the teacher to repeat answers when needed as well. Students could also talk to each other via typing comments in real-time. In total, the Live review session lasted about an hour, the time limit set by Instagram; however, at this point, students ran out of questions to ask, and the teacher reiterated final test reminders. In total, 113 users “attended” the Live review session, though the teacher only had 88 students (73 Honors Biology and 15 AP Biology). The additional attendees were former students and friends of current students in the two classes. There were students enrolled in the classes who did not attend the Live session for various reasons, including not wanting to spend study time on a Live session or not having access to Instagram at home.

5.3 | Data collection

5.3.1 | Online questionnaire

When students returned their assent and consent forms, they were given a personal identification code on a slip of paper and separately emailed a link to the *online questionnaire*. The questionnaire, which took no more than 25 min to complete, comprised of both semi-structured and open-ended questions. The purpose of the questionnaire was to receive feedback regarding the uses, limitations, and reactions to using Instagram. Additionally, demographic data about the participants was also part of the questionnaire. The girls had the option to opt-out of the study following completion of the questionnaire.

5.3.2 | Focus groups

Four *focus group* discussions were held with students who consented. All focus groups were kept small, with three participants per group, to maintain a productive conversation, and lasted about 30 min. The focus groups did not

extend more than an hour past dismissal time and took place in the school building. As an incentive to stay, a snack was provided. The girls provided feedback on the uses of Instagram in their science class, and they carefully considered, responded, and built upon the ideas and reactions of their classmates in the focus group, making it a valuable opportunity to collect data. All focus groups began with the same questions; however, due to the nature of the discussion, additional questions and comments were posed based upon student responses; therefore, no two focus groups were identical.

5.3.3 | Individual interviews

Students were invited to participate in an *individual interview*. These individual interviews took place two weeks after the Instagram Live review session. Overall, seven individual interviews took place. The individual interviews also allowed for a private setting and dialogue between the student and the teacher/researcher. They lasted 30 to 45 min and occurred within the school building during a time when students did not have a class to attend, or immediately following the last class of the day. In the individual interviews, students were able to provide greater detail regarding their reactions to the class Instagram account. The semi-structured interview questions were the same; however, the interviews were flexible. Each student responded personally based upon her experience of learning science with Instagram. For students who participated in the Live event, their interview included questions to gather feedback on the particular use of Instagram Live and how it helped with their exam performance. Both the focus group meetings and the individual interviews were audio-recorded and transcribed for analysis.

5.3.4 | Personal observations

Additionally, there were ongoing conversations during classes about how Instagram was being utilized. Students would spontaneously bring up ideas of how to further incorporate Instagram in academic ways, questions about specific posts, or even suggest posts based on things they had seen. Because of this, data collection also occurred constantly throughout the school year in the form of *personal observations*. By recording the brief discussions and comments made by students about the use of Instagram in the science classroom, we were able to garner a candid and evolving look at how students were reacting to its classroom implementation. Furthermore, by examining which online posts were most responded to by the students, both through in-class discussions and the “likes and comments” on the actual posts, we could assess the type of posts that had an impact on student learning.

Student use of direct messaging (DM) was also an important behavioral activity that needed to be observed throughout the study. For example, students would use the DM feature of Instagram to communicate with the teacher, ask questions, upload pictures for the teacher to post, and share ideas for future posts. Because students were already comfortably using DM to communicate with friends outside of class, this was an important way to see whether youth culture was being integrated with science learning. All communication from daily interactions was immediately transcribed into a field notebook and later used NVivo, a digital, qualitative data analysis software, to enter notes for data analysis.

5.4 | Data analysis

The authors referred to Creswell's (2007) suggestions of how to best analyze ethnographic information in which many data sources can be collected and compared to create a description of a culture-sharing group, in this case, Instagram users in a biology classroom. First, we began by compiling a detailed description of our culture-sharing group, which was the female biology students' participation in the use of Instagram for science learning.

All the questionnaires, focus groups, individual interviews, and observations were considered data to be transcribed, analyzed, categorized, and stored using NVivo software. Our coding process involved reading and re-reading the data to develop initial codes and writing memos. We identified codes and categories from the various sources of data by highlighting recurring words and phrases used by participants and designating them into specific groups that seemed to best describe the information we collected (Creswell, 2007). We saw patterns in the codes and categories and developed themes from the data. Here the themes were terms that arose from specific conversations and components of how Instagram and Instagram Live integrated youth culture in the science classroom.

By identifying these major themes in conversations with participants, additional information could then be found by locating these same concepts in the other data collected. As responses were compiled, it became clear that specific themes were emerging. Transcripts of conversations were juxtaposed with the questionnaires, allowing us to identify that participants were using Instagram to fill a specific need. Cross-case analysis permitted us to see how all the participants were using Instagram in class, how they were participating, and their implementation of Instagram during the Live review session. Students responded in ways that focused on how the use of Instagram in their class related to their identity as DNs and how it allowed them to view the course content in a new and different way. As a result of these responses, data were organized to highlight these emergent themes.

Lastly, the validity, reliability, and rigor of this study and its findings were measured using the guidelines provided by Lincoln and Guba (2007). The most useful was peer debriefing and discussing the data from two perspectives, as the first author was the teacher/researcher, and the second author assisted as a professional peer in the research design and interpretations. Data triangulation, which uses multiple forms of data collection, as we had done, and member-checking are valued strategies for ensuring validity in qualitative research. By comparing themes, evaluating responses of the 32 students who participated in the study, using member checking while completing the analysis and writing, and using rich-thick, description, these approaches ensured the rigor and analyses of the data and findings. In the following section, we discuss the two major themes from the data analysis process.

6 | FINDINGS

Overall, the study established a positive response to the use of Instagram in the science classroom from the female participants. Based on the qualitative responses, Instagram engaged students in the learning process more easily as the app was heavily aligned with their social media habits and allowed for additional class interaction that extended beyond the school day. Participants noted the ability of Instagram to bring youth culture into science learning by making a social habit now an academic one. The presentation of the findings is largely two themes that show how Instagram is used as an educational tool in science and the daily lives of female DNs.

6.1 | Theme 1: “Meeting Us Where We Are”

Student dependence on technology has increased a need for education to transform and fit into the habits of the DNs. Students were using Instagram frequently, so they encountered the classroom Instagram posts organically. According to student feedback, checking class websites and responding to email were not integrated as part of their daily routine, but using Instagram to serve those purposes ensured that students received messages from the teacher. During the group discussion, one student, Sutton said, “Teens check social media more often than their email or other class websites, so keep posting on Instagram” (Sutton, focus group). Furthermore, students reacted to using social media for incorporating their youth culture as digital learners into their science classroom experience. When asked if she felt the Instagram account was useful, Rebecca responded to how it was useful for DNs:

Yes, because a lot of what Ms. Serpagli posts on Instagram not only relates to what she teaches us, but it also relates to us in this generation. There

are educational memes or text posts or other things that are relatable to us not just because of what form they’re in but also because we learn a lot in biology. (Rebecca, questionnaire)

Similarly, Jazmin commented on how using social media was essential to her generation, the DNs. She expressed that technology was made for educational purposes:

Of course, it is, that’s what technology was made for, to be used for education. It enhances the classroom, keeps people awake and alert, and it is a constantly evolving thing that makes life easier. Especially as we get deeper into a generation that has not known life without technology, it is a way to stay relatable to the student to capture their attention in the classroom. (Jazmin, questionnaire)

During class one day, Ines remarked that Instagram was a useful way for students her age to connect to classroom content: “I think it’s great that you use Instagram. It meets us where we are. We use it, so it makes sense that you would use it for us” (Ines, personal communication). Her comment was unsolicited, and throughout the year, students voluntarily provided feedback about their reactions to Instagram, perhaps because they knew data were being collected regarding its implementation in the science classroom, or perhaps because they genuinely felt their remarks about using Instagram were helpful to share.

By using social media as a classroom tool, students were more likely to access it. For instance, during a focus group session, Mina discussed the convenience and ease of communication and contact and felt Instagram DM was more reliable than using email. Mina also felt that Instagram was approachable, more casual than email, and useful as a societal norm:

This is definitely a step up in terms of anything educational, having students take in information in the most convenient way and most approachable way since Instagram is so casual and like so embedded in our society like I would never imagine if I were a teacher to use such a colloquial way to have kids learn and, in a sense, it makes it so much easier. (Mina, focus group)

Mina also added how using Instagram was beneficial in learning science from using the digital tools embedded with the platform. She stated:

Freshman Bio could have been so much harder for everyone if like we were learning it the way some teachers in this school teach because Bio has a lot of information and it’s so much to

contain, but like, with little [videos of] dances, with funny memes, it was so much easier to understand. And I think because it was so casual, kids got it more and they were more inclined to put effort because it was approachable. (Mina, focus group)

In a different focus group, Antonia shared the same sentiments more concisely. She felt the use of Instagram was "...a really great idea" because she shared, "I'm always on Instagram and I think it's a good way to like connect to your teacher too" (Antonia, focus group).

Many of the girls who participated in the focus groups and interviews felt Instagram worked in the science classroom because students were on Instagram and always checking it. It worked through convenience and integrated easily into their technology habits. In her focus group, Antonia summed up saying:

I love using Instagram for school, for biology, I think it's great. Yea, I think it's a great way to relate, cuz it's like, you know, we're like young, and social media and everything, and like just bringing the classroom to us is really great. (Antonia, focus group)

For the DNs in this study, using Instagram in the biology classroom integrated well for educational purposes since the

students were already using it in their personal lives. The casual way that it was integrated into the biology classroom made the content easier to learn.

6.2 | Theme 2: "Having Class at Home"

Connecting to the previous theme, this second theme was related to ease and access to Instagram for learning science outside of the traditional classroom walls. A meme advertisement (Figure 1) was created and shared with the girls. They commented on the upcoming session and shared it with others. Students responded to a post that confirmed and reminded them of the Live event. Instagram "handles" (names) are covered by colored blocks to maintain confidentiality.

Four girls were asked to provide additional feedback on their reactions to the Instagram Live review session—Antonia, Ines, Marissa, and Theodora. Their reactions were recorded after the students took their midterms so they could adequately evaluate if it prepared them for the midterm. Ines and Marissa took part in individual interviews, while Antonia and Theodora were involved in the same focus group. When asked to provide feedback, the students all expressed positive reactions to Instagram Live. For example, Ines had originally suggested that the class try a Live session because more students would likely attend. After the Live session, she remarked, that it was a useful way to engage youth. She stated, "My brother follows you now. He finds your stuff helpful. He

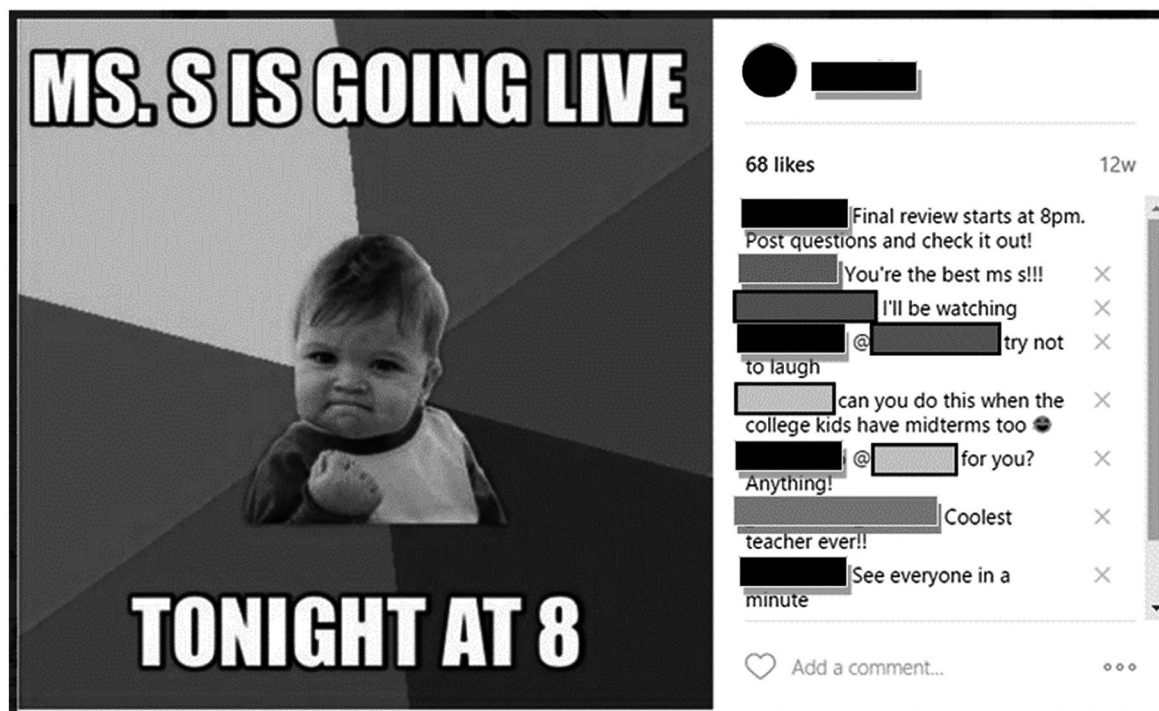


FIGURE 1 Instagram live announcement. Students responded to a post that confirms and reminds them of the LiveStream event. Instagram handles (names) are covered by colored blocks to maintain confidentiality

was on the Live. I was like, ‘Yea, I knew it would help everyone’” (Ines, Interview). Antonia said, “I thought it was a great idea. It was almost like a live classroom, so it was basically like having class at home, and I think we should have it more often.” Theodora shared her reactions to the Live session as “like class at home” which helped many students. “I liked it a lot because, like Antonia said, it was like class at home, and this class was really helpful, and it wasn't just helpful to us, like it helped...so many people were affected by the Live.” While wearing her pajamas at home in class, Theodora enjoyed the Live session as an opportunity to get her questions answered in preparation for the midterm:

We asked questions like you would in the classroom and it was that experience, but we were at home, in our pajamas, chillin', going over our notes and, honestly, I was freaking out before all my other midterms and I had, like, lists of questions and was like, “Well, can't get those ones answered” and then ... I was just typing [my biology questions] in the Live. And I was like I know this, and I felt good, it helped. (Theodora, focus group)

Many participants remarked that the Live session was efficient and helpful as they were able to review for a few hours by themselves before the review session began. This differs from in-person review sessions that occurred in the science classroom during or after school. In these settings, students have not always had the chance to review the material as thoroughly, leaving them with questions after the review with little opportunity to find answers. Theodora also remarked that she felt calm during the test, as the Live session prepared her for what to expect. However, not all the students were able to attend the Live review since one must have an account. To accommodate these students, a recording of the video was emailed to all students and posted to the course website so that all students had access to it after the Live took place.

Using Instagram Live for female students' participation in science and the issue of gender inequity in Science, Technology, Engineering, Mathematics, and Computer Science (STEM-CS) became a topic of discussion with the four girls who participated in the interview and focus group about the Live session because some students from the two courses invited their friends who attended other schools to join the Live review. Students from two neighboring all-boys high schools and a private co-ed high school also logged on and asked biology-related questions for their midterm that was scheduled on the same date as our test. A question in the interviews that came up centered around if they felt the experience was tainted or less effective because “outside” students had joined the Live review session. The students seemed to have a great deal of pride that they were using technology

in their all-female institution in such an innovative and unusual way of learning science. They wanted other students to not only see what they were doing but to also share in that experience. Consequently, students asked to continue using Instagram Live in their biology class. The participants in the study felt that it provided an additional opportunity to interact with each other, their teacher, and other friends outside of their school. Thus, the Live science review session was a positive experience that integrated technology, teaching and learning, and science in an all-girls school. The opportunity to learn at home and to use an innovative approach to science learning was beneficial to the participants (and their friends and siblings) through this interactive social media platform.

7 | DISCUSSION OF FINDINGS

The findings of this study consider how Instagram as a digital tool brings science into the daily lives of female high school DNs. Using Instagram supported their out-of-class learning and increased access for girls in science learning by meeting them where they are. First, using Instagram as a classroom tool was “casual” to the students because they were already using it in their daily lives. Instagram incorporated their youth culture and their interests in social media as a tool for learning and studying science, which also created an atmosphere of comfort and relatability. Because these students are classified as DNs (Prensky, 2001) based on their digital comforts, how they use digital media, and their ability to comfortably integrate digital media into many aspects of their lives, their openness to trying social media as a classroom tool supported their science learning. Students almost all agreed that the visual cues and responses they received through comments, posts, and DM conversations were useful and successful ways to bring their youth culture into the science classroom, making science learning more understandable, desirable, and participatory. The Instagram Live happened because the girls in the biology class were using Instagram and its features and asked to use Live as a method for review. Students also guided the Live event because they typed the questions, responded to each other, and pressed the heart icon to “like” a post to indicate their understanding. Most of the girls remarked in their questionnaire their comfort with Instagram, and their habitual use allowed a seamless integration. They were accessing the biology Instagram more frequently than other educational resources, mostly because it is a platform that complemented their DN identity. Students were actively engaged in the learning process, while also building a digital community of practice.

Second, the Live online review session, which lasted one hour, replaced the traditional in-school midterm review session. Participating in the Live setting at home was more relaxed than a traditional classroom, allowing students to feel

more at ease with the process of learning in the comfort of their home. During the Live session, girls had the opportunity to ask questions, request to repeat answers, and receive test hints and reminders. The students could join or leave the Live review at any time during the session. The session was recorded so that students who could not participate or those who did and wanted a refresher could watch later. Thus, the Live session allowed more flexibility and increased participation for students. It became a biology discussion and review where students conversed with each other about academics in a non-academic setting. This was shown in the number of students who participated, including male siblings and friends from other schools. Following the exam, students remarked that they felt prepared, knowledgeable, and fluent in the topics covered based on the previous night's Live.

Third, it was possible to qualitatively gauge how the many uses of Instagram in this science classroom impacted student engagement in a positive way. The girls were communicating with their teacher outside of school through the social media platform. Students who did not ask questions often in class were leaving comments and asking for clarification. Many students were clicking on the heart icon to "like" a post, which affirmed that the post reached that student, and could even indicate appreciation of the post or acknowledge understanding. Overall, this signified that students were actively engaged with the digital conversation. Instagram aided content awareness and students' active participation through likes, DM, and comments. A future study could potentially juxtapose various social media platforms to determine which acts as the most useful tool to augment classroom learning. Investigating how these social tools could be further used, such as in an event of a stay-at-home-order could also provide greater insight into the growing needs of keeping up with the digital and cultural trends of DN students.

8 | IMPLICATIONS

As an educational tool, Instagram offered an opportunity to reach female students in science through a nontraditional pedagogical approach. Instagram, as a channel for learning, made science relevant, innovative, and digitally accessible for the participants in this study. It allowed these female science students to be exposed to STEM-CS on a personal level. The girls in the study continued to bring up how impressed boys from nearby schools were regarding the class's use of social media in the science classroom. Social media has the potential to alter the views of young women regarding STEM-CS learning by incorporating a familiar app into their science learning. Their responses to its uses in-school and out-of-school as well as their encouragement to continue using the Instagram account as an educational tool for science learning were welcomed. Feedback and comments via

social media offered a positive assessment that also raised students' self-esteem and provided opportunities for recognizing student voice, which is often missed in the formal, structured patterns of digital media use (Clark et al., 2015). As mentioned earlier, providing female students with support and outlets for curiosity in STEM-CS can help to narrow the STEM-CS gender gap (Brotman & Mensah, 2008; Malmimi, 2009; Mensah & Serpagli, 2020).

The Live session and the feedback the female students received from their male peers encouraged the "growth mindset" that Hill et al. (2010) mention: "...believing in the potential for intellectual growth, in and of itself, improves outcomes. This is true for all students, but it is particularly helpful for ... girls' achievement and interest in math and science" (p. xiv). This experience was not taking place in their all-female school but outside, with male counterparts and other outside participants, allowing the girls to show confidence for their participation in science with the use of digital tools for learning. They disregarded the negative stereotypes that persist in STEM-CS and invited male students to attend, ignoring the intimidation of having males "compete" in this online setting. Instead, they shared online space, making it more inclusive. The girls were not concerned with how their questions would be perceived; they simply thought that they had access to a more exciting and accessible space to learn. The girls wanted others to see and to experience learning science in a new way: a digital pedagogical tool for DNs like them.

All students, and especially girls, must be provided with examples and outlets that allow them to experience as much science learning success as possible (Mensah & Serpagli, 2020). Instagram is just one of the many ways that gender gaps can be combated, refuted, challenged, and contested when social media is used for educational purposes. Social media is an updated, digital channel through which teachers can create and maintain relationships. Communication is no longer through face-to-face interactions, phone calls, or even emails; people can also use private messages, comments, and likes on posts in the digital world. These digital innovations have produced an arena for individuals to be seen and heard, and DNs have taken to digital tools as part of their daily lives. Thus, educators are at a unique point where we can look to see how different forms of digital communication can benefit and extend our classroom instruction in ways that academic information is both useful and relevant to our students. Even in current times of remote teaching and learning due to the COVID-19 global pandemic, social media brings a DN-centered approach to support student learning (Hodges et al., 2020).

We must also consider that social media is constantly in flux and that not all students have access to internet and digital tools (Gonzales et al., 2020). The popularity of various apps and platforms is always changing, and new apps are

always emerging. The ebb and flow of which social media apps are best utilized in a classroom must remain flexible. Teachers can choose which and how they use social media and other digital formats to best meet the needs of their learners, their teaching, and content. By presenting how Instagram worked specifically in the biology classroom, those same ideas can be generalized and used for a greater understanding of how social media can enhance digital literacy (Jones-Kavalier & Flannigan, 2006) and classroom learning that meets the needs of our DN students, not just in the science classroom but also in other subject areas. Overall, the participants expressed that as many students as possible should have similar opportunities to integrate digital technologies into education.

9 | CONCLUSION

By using Instagram Live as an innovative technological and educational tool in the science classroom, students at an all-girls school digitally engaged in science learning. Using social media, a familiar tool with these female DNs, made accessing biology content consistent with their interests, habits, and youth culture. The Live review session provided an instance in which the girls were able to communicate with their classmates, teacher, and others in learning science. The online review drew upon the traditional in-class study sessions, with questions and responses, but without the need for the physical institution and traditional hours of engagement. The participants were comfortable and seemed pleased to tell friends they were using a popular social media app in their science classroom that other schools and students were not yet using for educational purposes, and certainly not in the context of science learning.

We must recognize that technology can both inhibit and empower girls in the classroom by creating arbitrary gender roles and discouraging growth in all aspects of the STEM-CS fields. The hope is that this study has found one of many ways to open doors for students to continue upon a path of growth, understanding, and achievement in science by incorporating both their need for constant classroom access as well as acknowledgment of their youth culture in a classroom setting. By finding ways to instruct that is culturally responsive, while still challenging and appropriate, DNs will remain engaged and excited about learning.

CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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